

Project Title:

Retail Sales Analytics Dashboard

1 Business Objective

Analyse sales, profit, customer behaviour, and regional performance to help management make better business decisions.

2 Dataset Overview

- Source: Kaggle
- Rows: 9,994
- Columns: 21

3 Data Quality Assessment

- Missing Values: None
- Empty Values: None
- Errors: None
- Duplicate Rows: None
- Data Types:
 - Order Date → Date
 - Ship Date → Date
 - Sales → Decimal Number
 - Profit → Decimal Number
 - Discount → Decimal Number
 - Quantity → Whole Number
 - Postal Code → Text
 - Remaining descriptive fields → Text
- Remarks:

Dataset contained no missing values, duplicate rows, or data errors. Data types were validated and corrected before loading into Power BI.

4 Data Cleaning Performed

Transformation	Reason
Converted Order Date to Date	Enable time-based analysis
Converted Ship Date to Date	Enable time-based analysis
Converted Sales, Profit, Discount to Decimal	Correct numerical calculations
Converted Quantity to Whole Number	Represents count of items
Converted Postal Code to Text	Identifier, not a numeric measure
Removed Row ID	No analytical value

5 Data Model

- Imported as a single fact table model.
- No relationships were required.
- A dedicated Measures table was created to organize DAX measures separately from the dataset.
- The model was intentionally kept simple because the dataset consists of a single transactional table.

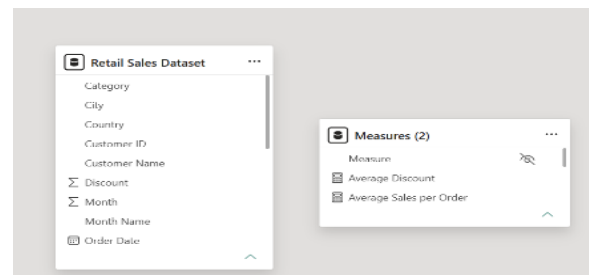


Figure 1: Data Model

6 DAX Measures

Measure	Formula
Total Sales	SUM(Sales)
Total Profit	SUM(Profit)
No. of Orders	DISTINCTCOUNT(Order ID)
Average Sales per Order	DIVIDE([Total Sales],[No. of Orders])
Total Quantity	SUM(Quantity)
Average Discount	AVERAGE(Discount)
Profit Margin	DIVIDE([Total Profit],[Total Sales])

7 Power BI Dashboard

7.1 Retail Sales Executive Dashboard

The Retail Sales Executive Dashboard provides a high-level overview of the company's sales performance from 2014 to 2017 using KPI cards and interactive visualisations. It summarises key business metrics, monthly sales trends, regional performance, category-level sales, and profitability while allowing users to explore the data through interactive slicers.

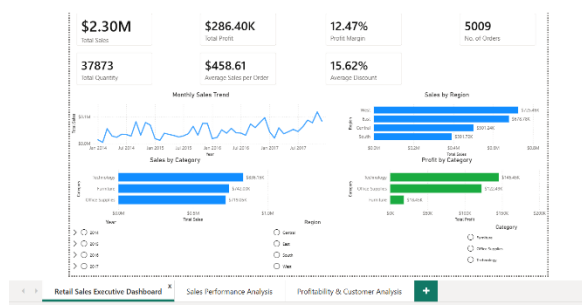


Figure 3: Retail Sales Executive Dashboard

7.2 Sales Performance Analysis

The Sales Performance Analysis dashboard evaluates sales and profitability across product categories and sub-categories. It highlights differences between revenue generation and profit contribution and enables the identification of high-performing and underperforming product groups.



Figure 4: Sales Performance Analysis

7.3 Profitability and Customer Analysis

The Profitability and Customer Analysis dashboard examines profit margins, loss-making states, high-value customers, and discount patterns across product sub-categories. The dashboard supports deeper investigation of factors influencing profitability and areas requiring management attention.

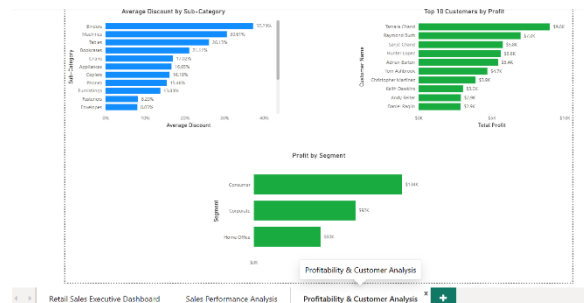


Figure 5: Profitability and Customer Analysis

8 Business Insights

- The company generated approximately **\$2.30 million in sales** and **\$286K in profit** from **5,009 unique orders** during 2014–2017.
- Monthly sales generally increased over time, with the highest monthly sales occurring in **November 2017**.
- The **West region** generated the highest sales, while the **South region** generated the lowest.
- **Technology** was the strongest revenue-generating category and achieved the highest overall profit.
- **Furniture** generated substantial sales but had a significantly lower profit margin, indicating that high sales do not necessarily translate into strong profitability.
- **Labels** and **Paper** achieved the highest profit margins, while **Supplies, Bookcases, and Tables** recorded negative profit margins.
- **Tamara Chand, Raymond Buch, and Sanjit Chand** were among the highest-value customers based on profit contribution.
- **Texas** was the worst-performing state by profit, followed by **Ohio and Pennsylvania**.
- **Binders** received the highest average discount while maintaining a positive profit margin. **Tables and Bookcases** received relatively high discounts and generated negative margins.
- Discount level alone does not fully explain profitability, suggesting that pricing, product costs, customer behaviour, and operational factors should also be considered.

9 Business Recommendations

- Investigate the Furniture category and its low-margin sub-categories to identify pricing, discount, product cost, and operational factors affecting profitability.
- Review the performance of Supplies, Bookcases, and Tables and develop targeted strategies to reduce losses.
- Analyse high-discount transactions alongside product costs and purchasing behaviour rather than evaluating discount levels in isolation.
- Study the purchasing patterns of high-value customers and develop targeted retention or loyalty strategies.
- Investigate loss-making states, particularly Texas, Ohio, and Pennsylvania, with attention to discounting, product mix, and logistics.
- Use historical monthly sales patterns for demand forecasting, inventory planning, and marketing preparation ahead of peak periods.
- Apply targeted customer engagement and promotional strategies during historically weaker sales periods.